

Project 1: Simon's Algorithm

```
[1]: import numpy as np
```

```
[2]: from qiskit                import QuantumCircuit, transpile
      from qiskit.visualization import plot_histogram

      from qiskit_aer           import AerSimulator
```

1 Oracle Gate

```
[3]: def s_oracle(b, n):
      oracle_qc = QuantumCircuit(2*n)
      b = b[::-1]
      for q in range(n):
          oracle_qc.cx(q, n + q)
      for q in range(n):
          if(b[q] == '1'):
              x = q
              break
      for q in range(n):
          if(b[q] == '1'):
              oracle_qc.cx(x, n + q)
      # convert circuit to gate
      oracle_gate = oracle_qc.to_gate()
      oracle_gate.name = "Oracle"
      return oracle_gate
```

2 Algorithm Circuit

```
[4]: def s_algorithm(oracle, n):
      s_circuit = QuantumCircuit(2*n, n)
      # Initial Hadamard
      for qubit in range(n):
          s_circuit.h(qubit)
      # Adding Oracle
      s_circuit.append(oracle, range(2*n))
      # Final Hadamard
      for qubit in range(n):
          s_circuit.h(qubit)
      # Measurement
      for i in range(n):
          s_circuit.measure(i, i)
      return s_circuit
```

3 Create Circuit

```
[5]: n = 3
b = '110'
oracle_gate = s_oracle(b, n)
qc = s_algorithm(oracle_gate, n)
qc.draw('mpl')
```

[5]:

